

ES -116

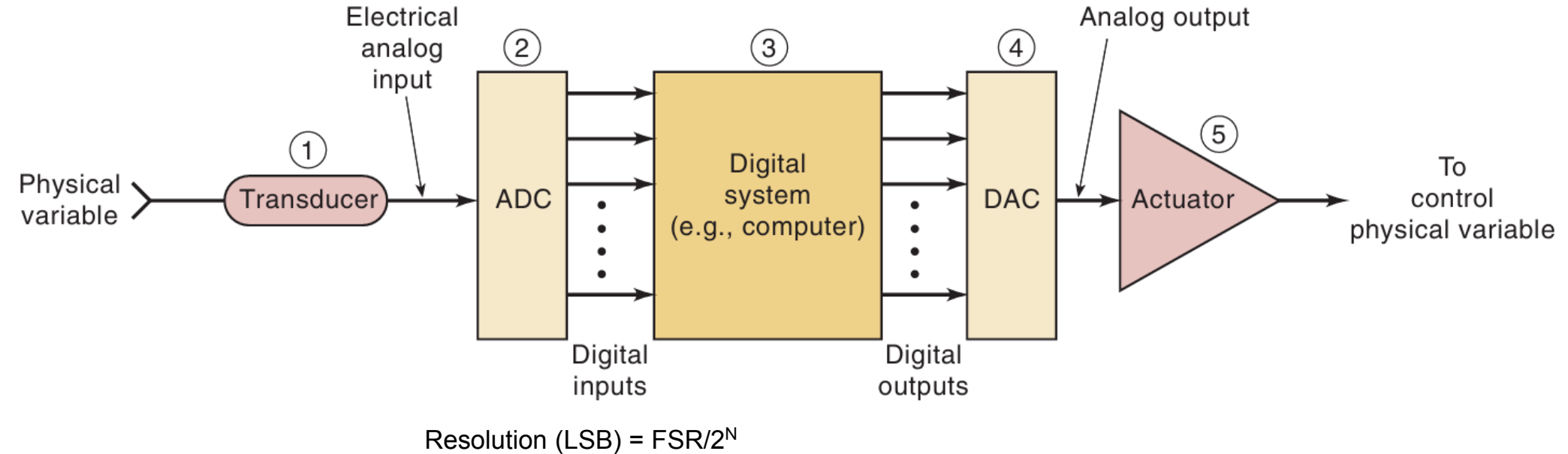
Principles and Applications of Electrical Engineering

Semester II, 2025-26

Entering the Digital World

Reference book: Chapter on “Interfacing with the Analog World” from Digital Systems Book by Tocci

Recap



Examples: DSO, Automotive Cruise Control, Autonomous “this and that” etc..

Warm-up Question

A sensor outputs a voltage in the range 0 V to 8 V. You are designing electronics for different applications.

Which of the following applications can be implemented using only a 1-bit ADC (comparator), and which require multiple bits?

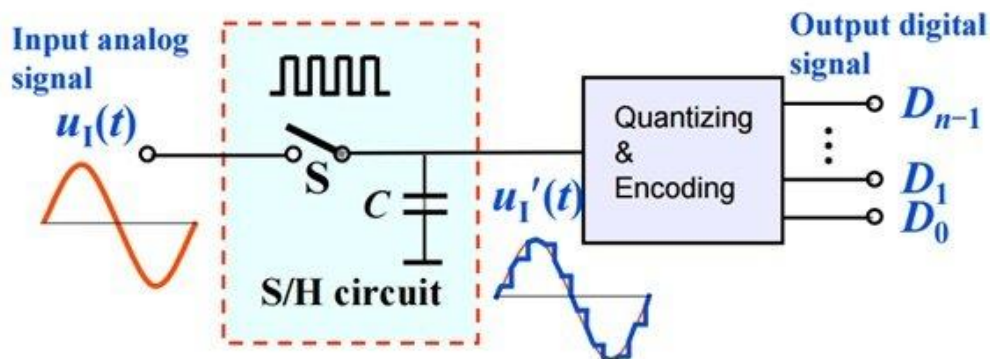
Case 1: Turn ON a cooling fan only if temperature exceeds a safe limit

Case 2: Display the exact temperature value on a screen

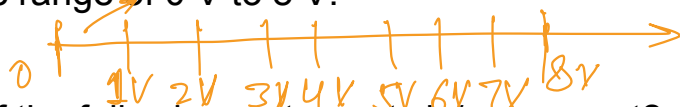
Case 3: Trigger an alarm if gas concentration is above a threshold

Case 4: Control motor speed smoothly based on sensor value

Candy Question

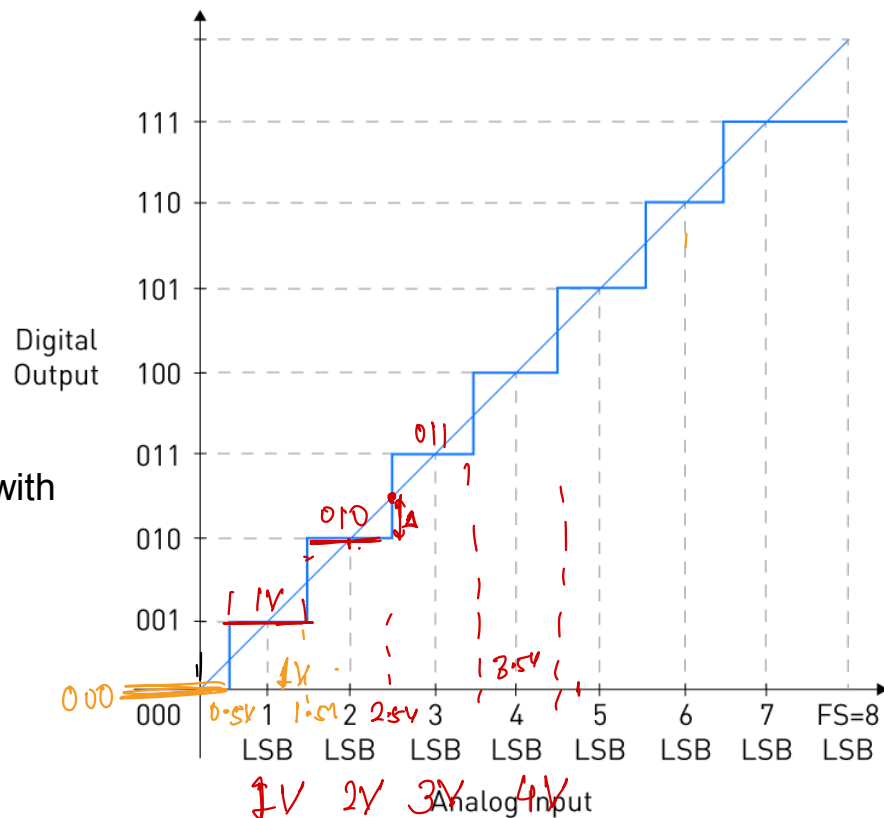


The figure shows the input-output characteristic of a 3-bit ADC with full-scale range of 0 V to 8 V.

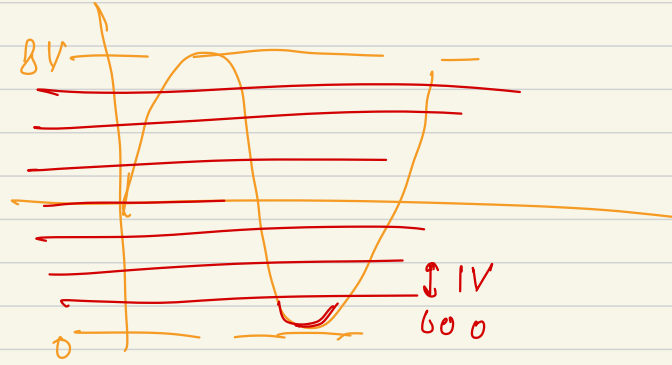


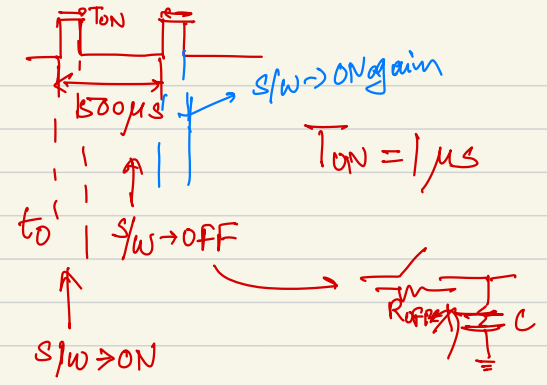
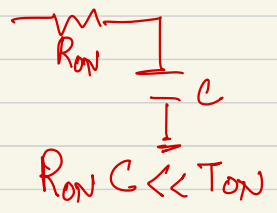
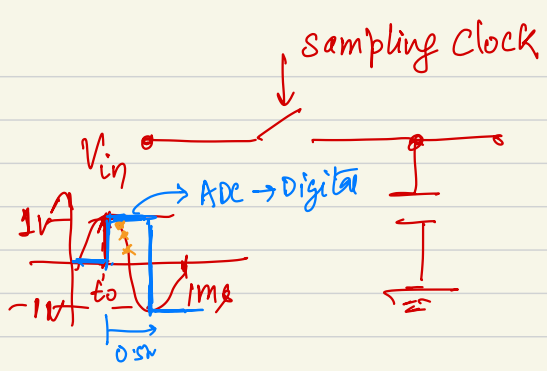
Which of the following statements is/are correct?

- A. The resolution (LSB size) of the ADC is 1 V ✓
- B. An input of 4.7 V will produce the digital output 100
- C. An input of 4.7 V will produce the digital output 101
- D. Inputs between 5 V and 6 V produce the same digital output



$$LSB = \frac{FSR}{2^n} = \frac{8}{8}$$





one design of switch

control signal

high

R

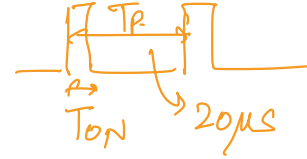
$R \rightarrow low [V_c: high]$

$R \rightarrow high [V_c: low]$

Sample and Hold.....

An ADC uses a sample-and-hold (S/H) circuit. The system samples the input every $20\ \mu\text{s}$, and the ADC requires $8\ \mu\text{s}$ to complete each conversion.

The input signal is continuously varying.



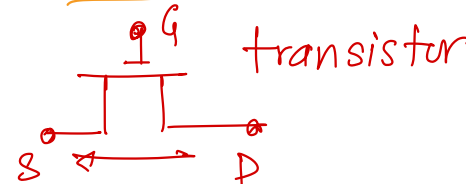
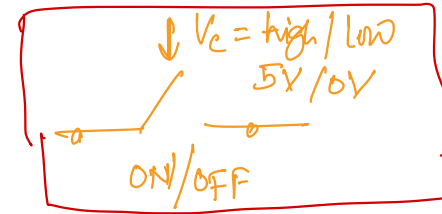
Which of the following statements is/are correct?

- A. The input is sampled at discrete instants separated by $20\ \mu$
- B. After each sampling instant, the S/H holds the sampled value constant for the next $20\ \mu\text{s}$
- C. The ADC converts the value of the input signal at the sampling instant
- D. During the $8\ \mu\text{s}$ conversion time, the input to the ADC remains constant due to the hold operation
- E. If the input signal changes during the hold period, it affects the current conversion result

In a sample-and-hold circuit, a switch is controlled by a clock signal.

Which statement best describes the role of this switch?

- A. It amplifies the input signal
- B. It allows current to flow only when a control signal is applied
- C. It converts analog signals to digital signals
- D. It permanently stores the input voltage



Is ADC loosing information?

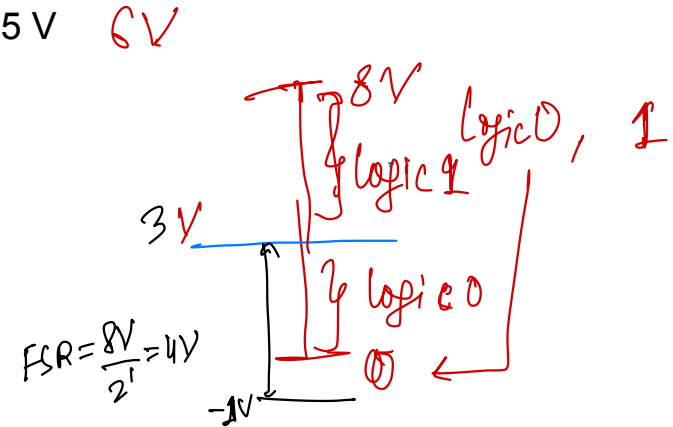
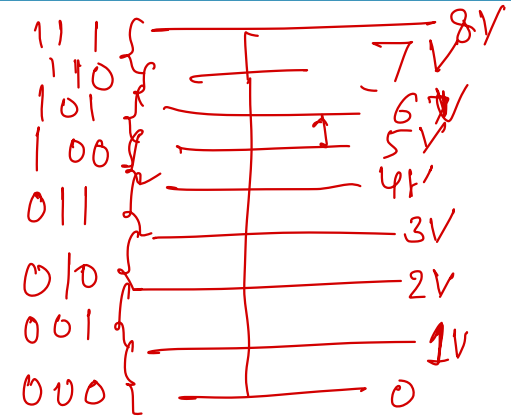
A sensor outputs a voltage in the range 0 V to 8 V. Two ADCs are available:

- ADC 1: 3-bit resolution
- ADC 2: 1-bit resolution with reference voltage $V_{\text{ref}} = 4 \text{ V}$

The input signal is 5.2 V.

Which of the following statements is/are correct?

- A. The 3-bit ADC will output the digital code corresponding to approximately 5 V
- B. The resolution of the 3-bit ADC is 1 V
- C. The 1-bit ADC will output logic HIGH
- D. The quantization error of the 3-bit ADC is less than that of the 1-bit ADC
- E. The 1-bit ADC can distinguish between inputs of 5.2 V and 7.8 V



Candy Question

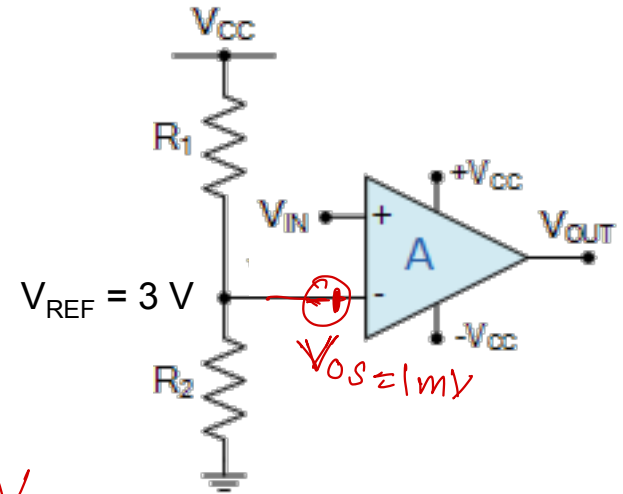
An op-amp is used in open-loop configuration as a comparator.

The non-inverting terminal is connected to a reference voltage of **3 V**, and the inverting terminal is connected to an input signal $V_{in}(t)$. The op-amp is powered by $\pm 15\text{ V}$.

To interface with a microcontroller, the output is interpreted as: $+15\text{ V} \rightarrow \text{Logic 1}$ and $-15\text{ V} \rightarrow \text{Logic 0}$

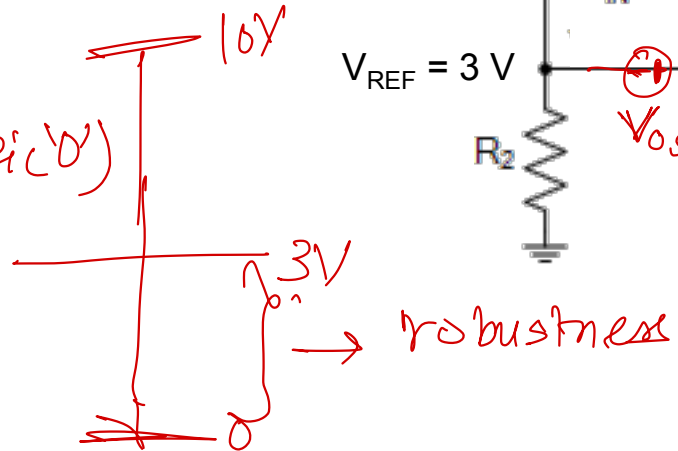
Which of the following statements is/are correct?

- A. The circuit behaves as a 1-bit ADC with threshold at 3 V
- B. If $V_{in} = 2\text{ V}$, the digital output is 1
- C. If $V_{in} = 5\text{ V}$, the digital output is 0
- D. The circuit converts a continuous input voltage into one of two digital levels
- E. The circuit can distinguish between 4 V and 10 V inputs



$$0 \leq V_{in} < 3\text{ V} \rightarrow V_{out} = -15\text{ V} (\text{logic '0'})$$

$$V_{in} > 3\text{ V} \rightarrow V_{out} = +15\text{ V} (\text{logic '1'})$$



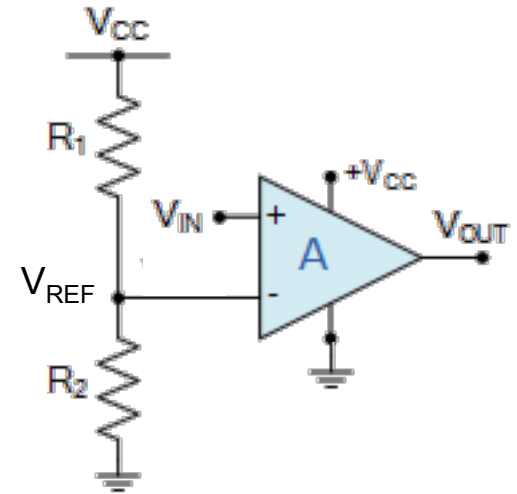
Check your understanding

An op-amp is used as a comparator with output interpreted as: $+5\text{ V} \rightarrow \text{Logic 1}$ and $0\text{ V} \rightarrow \text{Logic 0}$. (i.e. Op-Amp powered by $V_{CC} = +5\text{ V}$ and 0 V)

The inverting terminal is connected to the input V_{in} , and the non-inverting terminal is connected to a reference voltage V_{ref} .

Which of the following statements is/are correct?

- A. Changing V_{ref} changes the voltage at which the output switches between 0 and 1
- B. If $V_{ref} = 5\text{ V}$, the circuit always outputs 0 for inputs between 0 and 5 V
- C. If $V_{ref} = 0\text{ V}$, the circuit behaves like a detector for positive voltages
- D. The circuit converts an analog input into a 1-bit digital output
- E. Increasing the supply voltage changes the switching threshold



Candy Question

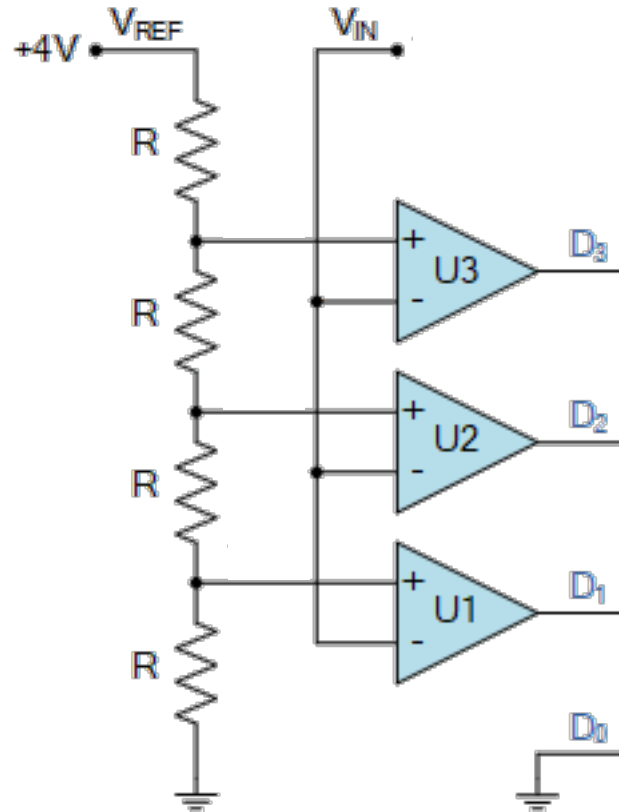
You want to design an ADC for an input range of 0 V to 4 V that produces 2-bit digital output using comparators. Op-Amp are powered by $V_{CC} = +5\text{ V}$ and 0 V).

Each comparator compares V_{in} with a fixed reference voltage and outputs: 1 (5 V) if $V_{in} > V_{ref}$ and 0 (0 V) otherwise.

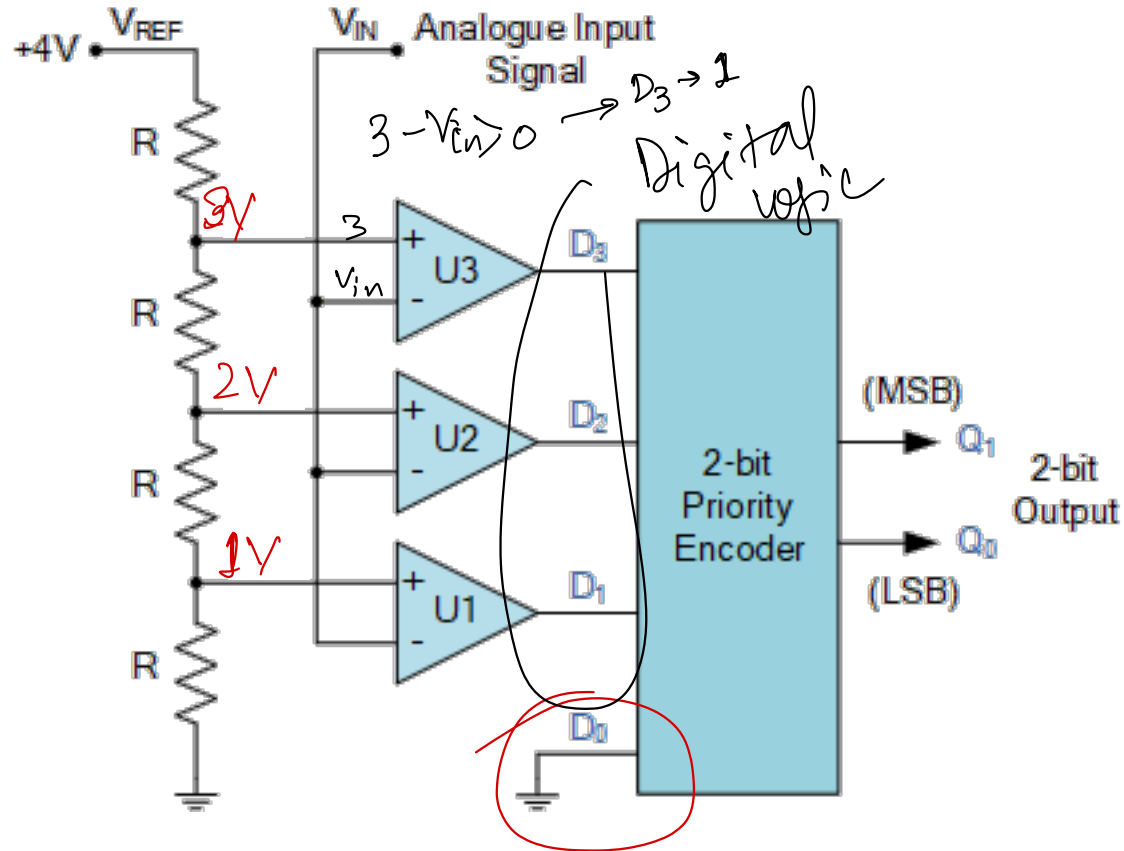
Which of the following design choices will correctly implement this ADC?

- A. Use 2 comparators with reference voltages at 1 V and 3 V
- B. Use 3 comparators with reference voltages at 1 V, 2 V, and 3 V
- C. Use 4 comparators with reference voltages at 0 V, 1 V, 2 V, and 3 V
- D. Use 1 comparator and vary its reference voltage dynamically

2-bit Flash ADC



2-bit Flash ADC



Analog Input Voltage (V_{IN})	Comparator Outputs				Digital Outputs	
	D_3	D_2	D_1	D_0	Q_1	Q_0
<u>0 to 1V</u>	1	1	1	0	0	0
1 to 2V	1	1	0	0	0	1
2 to 3V	1	0	0	0	1	0
3 to 4V	0	0	0	0	1	1

Why can't the comparator outputs be directly used as a 2-bit digital number?

- A. It has more bits than required
- B. It does not represent a unique binary value
- C. It is not stable
- D. It requires amplification

Before we move away from the Analog world...

A 2-bit ADC operates over the range 0 V to 8 V.

The ADC produces the digital output: 10

What is the corresponding analog input voltage?

- A. Exactly 4 V
- B. Exactly 5 V
- C. Any value between 4 V and 6 V
- D. Cannot be determined without additional information

“Bits don’t carry meaning, systems assign meaning to bits.”

What systems do we have in Digital World?

Number systems = language of representation

Binary (base-2)

- Natural for hardware
- But long and hard for humans

Example: $25_{10} = 11001_2$

Decimal (base-10)

- Natural for humans
- Not for hardware

Application: Displaying sensor values (temperature, voltage, speed), User inputs (keypads, GUIs)

Hexadecimal (base-16)

- Compact way to represent binary
- Easy for engineers

Example: $11001_2 = 19_{16}$

Hex is how engineers read binary without going crazy. Applications: Memory Addresses (e.g. 0x1A3F), Color codes (#FF0000), etc.

Why do digital systems use binary representation?

- A. It is easier for humans to understand
- B. Electronic devices naturally operate in two states
- C. It reduces power consumption
- D. It increases accuracy

Write $FACE_{16}$ in Binary

Swag of the Transistor

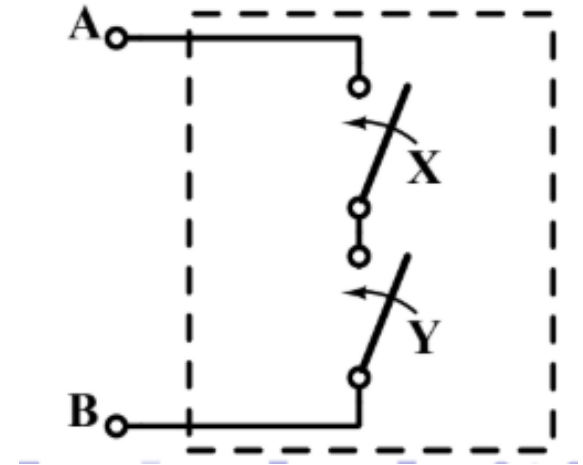
A device behaves like this:

- Control signal HIGH → switch ON → current flows
- Control signal LOW → switch OFF → no current flows

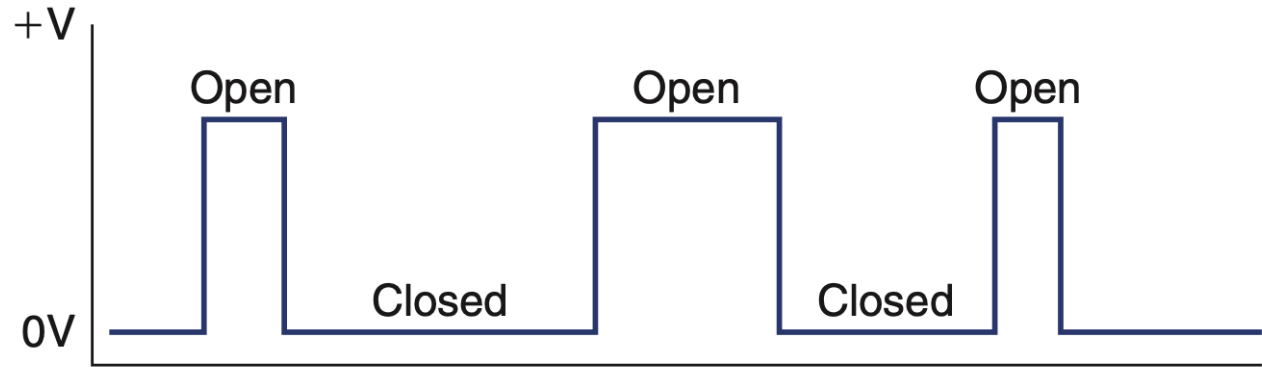
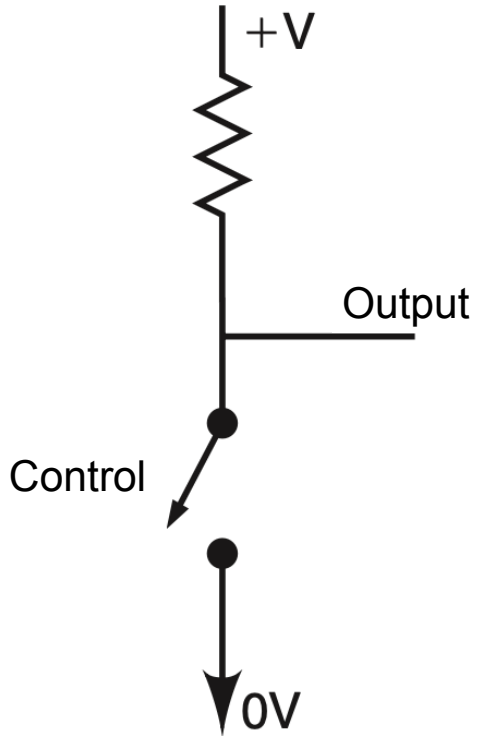
Why is such a device fundamental to digital electronics?

- A. It can produce continuous voltage levels
- B. It can represent two distinct states (ON/OFF)
- C. It increases signal strength
- D. It stores energy efficiently

Which
logic/functionality
does it make?



A simple Logical Operation



For the given Output signal, draw the corresponding Control signal for the switch.